Monitoring Chlorophyll a Concentration in the Mar Menor Coastal Lagoon Using Ocean Color Sensors

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Abstract

The Mar Menor coastal lagoon has experienced a severe eutrophication process since 2015, with its most visible effect being a series of phytoplankton blooms featured by unprecedentedly high and persistent concentrations of chlorophyll *a* in the water column. To better quantify and monitor these changes, two suitable chlorophyll concentration algorithms (referred to as BELA, short for BELich Algorithm) for the Mar Menor lagoon are proposed. These algorithms are derived from the analysis of thousands of reflectance spectra obtained from six satellite ocean color sensors, combined with *in situ* observations performed between 2016 and 2023. Our results demonstrate that the BELA algorithms can accurately estimate chlorophyll *a* concentration in the shallow waters of the Mar Menor lagoon for values above 2 mg m^{-3}, a range consistent with various of the episodes of high productivity observed since 2015. These algorithms perform well for a variety of currently operational ocean color sensors and could be used in future planned missions. This versatility makes them a valuable tool for the monitoring and assessing the environmental state of the lagoon.

1. Introduction

The Mar Menor coastal lagoon, one of the largest hypersaline lagoons in the Mediterranean Sea with an extension of 135 km^2, is increasingly threatened by anthropogenic pressures leading to nutrient pollution. Its elevated salinity (over 40), caused by intense evaporation that is not compensated by precipitation or freshwater inputs from the mainland, along with its shallow depth (not exceeding 7 m), creates a unique coastal ecosystem with a high ecological value that hosts a wide variety of species. This ecosystem suffered a severe dystrophic crisis at the end of 2015 (Mercado et al., 2021), probably triggered by decades of continuous discharges of nutrients coming from the polluted groundwater and other anthropogenic sources (Velasco et al., 2006;

Caballero-Pedraza et al., 2015). Since late 2015, recurrent phytoplankton blooms have led to the degradation of the entire ecosystem, including the disappearance of the macrophyte communities, anoxic events, massive deaths of aquatic animals, and whiting phenomena (Ruiz et al., 2022; Ouaissa et al., 2023; Gómez-Jakobsen et al., 2024). The first phytoplankton bloom documented in 2015 (the so-called "green soup" due to the intense coloration of the water) persisted from autumn of 2015 to autumn of 2016. During this period, chlorophyll *a* concentrations (Chla) exceeded 10 mg m^{-3}, levels never before registered in the lagoon. This bloom led to the collapse of benthic communities, which were subsequently replaced by communities of a different composition (Belando et al., 2019, 2024; Jiménez-Casero, 2024). Subsequent episodes of high productivity, lasting several weeks to months, were recorded in the summers of 2017 and 2019. During the latter event of high productivity, Chla concentrations over 20 mg m^{-3} were reached, leading to anoxia near the bottom and the massive death of aquatic animals (Ruiz et al., 2020). The most recent high productivity episode occurred in summer 2021, also resulting in extensive aquatic fauna mortality (Ruiz et al., 2022). Since then, Chla have remained below 2-3 mg m^{-3}, which are nevertheless high values compared to those registered before 2015 (typically not exceeding 1 mg m^{-3}; Ruiz et al., 2020). Additionally, in the spring of 2022, an area of white and turbid water appeared in the western part of the lagoon, probably linked to the ongoing eutrophication process (Gómez-Jakobsen et al., 2024).

The primary symptom of eutrophication is a burst in phytoplankton biomass, which significantly modifies the optical properties of the water column. Therefore, Chla has become a key indicator for monitoring the ecological status of the Mar Menor since the onset of the dystrophic crisis in 2015. However, previous data of Chla obtained in the lagoon are scarce and limited to sporadic research surveys, which hampers comparing the present chlorophyll dynamics with periods previously characterized by a higher ecological status, when Chla rarely exceeded 1 mg m^{-3}. Alternatively, the analysis of images captured by satellites provides a valuable resource for reconstructing time series of chlorophyll concentration. However, retrieving data of Chla from satellite images in the Mar Menor is challenging due to its shallow depth, as bottom reflectance can contaminate the signal captured by the satellite, especially when water transparency is high. This is likely the reason why most algorithms based on blue-green band reflectance ratios fail to provide accurate estimations of chlorophyll concentration in the lagoon, particularly when blue radiation reaches the bottom (O'Reilly & Werdell, 2019). Furthermore, Mercado et al. (2021) noticed that changes in phytoplankton composition could affect the reflectivity of the lagoon water. Other algorithms, based on the NIR (Near Infrared) bands (Gons et al., 2002), which perform adequately in highly turbid estuaries and river mouths with high loads of suspended sediments (Case 2 waters) could be used. However, their applicability in the Mar Menor is limited, as the contribution of this component to the lagoon's optical properties is relevant only during rare torrential rainfall events followed by runoffs of short duration (Velasco et al., 2006; Álvarez-Rogel et al., 2020), and are not related with higher Chla values. Additionally, NIR bands are not available in all current ocean color sensors.

Recent efforts have focused on developing empirical satellite algorithms to estimate Chla in the Mar Menor lagoon using multispectral images from the OLI-Landsat-8 and MSI-Sentinel-2 satellites (Erena et al., 2019; Caballero et al., 2022). However, the low temporal resolution of these remote sensing platforms may be insufficient to capture the rapid changes in the optical properties driven by the dynamics of the lagoon's ecosystem. For this reason, different authors have proposed developing algorithms based on ocean color platform data with higher temporal resolution (virtually daily) and combining data from multiple sensors that are available nowadays and provide equivalent observational information of the water's optical properties (Giménez et al., 2024). However, these approaches are limited by the amount of available *in situ* chlorophyll registers, which are necessary to build these algorithms. Moreover, these *in situ* Chla data are typically obtained by different methods, including fluorescence probes deployed directly in the water column and spectrophotometric measurements performed in the lab from water samples collected in the lagoon.

The main objective of this work is to design an empirical algorithm based on data from multiple ocean color sensors, to accurately estimate Chla concentrations in the Mar Menor lagoon by using the entire available *in situ* dataset.

2. Material and methods

The *in situ* chlorophyll *a* concentration data used for the development of the algorithm were gathered through two monitoring programs conducted by the Spanish Institute of Oceanography (IEO-CSIC) and the Regional Government of Murcia (CARM). The IEO-CSIC dataset consists of concentration measurements from samples collected at fixed stations between June-2016 and June-2023. Initially, six stations were sampled until May 2018, after which sampling continued at three of these stations (https://www.ieo.es/es/informes-sobre-la-evolucion-del-ecosistema-lagunar, Figure 1). Water samples were collected by triplicate at a depth of 4 m at each sampling station. The samples were filtered using GF/F Whatman filters, and the analysis of Chla was conducted by spectrophotometry, after extraction in 90% acetone overnight at 4°C, using the Lorenzen's method (Lorezen & Jeffrey, 1980).

CARM´s data were obtained weekly at 12 sampling stations using a calibrated CTD fluorescence probe deployed from the surface to the bottom (https://marmenor.upct.es/). For our analyses, the vertical profiles were averaged to produce a single Chla value for each sampling date and station.

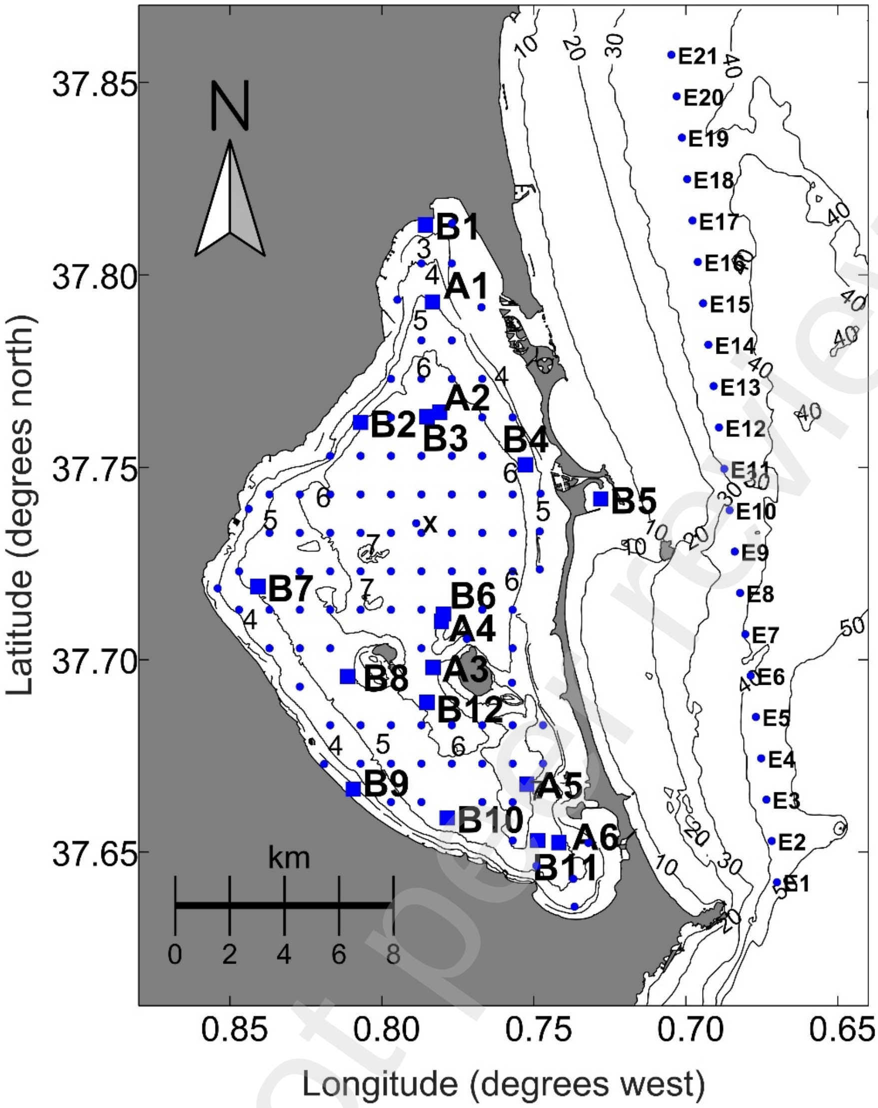

Figure 1. Map of the Mar Menor lagoon showing the sampling stations of the IEO-CSIC (A1-A6) and CARM (B1-B12), as well as the satellite sampling nodes (blue dots). Isobaths are labeled (m). x is the location of the center of the lagoon and E1-E21 are the satellite sampling nodes outside the lagoon.

The observation of chlorophyll concentrations obtained from the two monitoring programs indicates that both methods to estimate chlorophyll concentration reproduce adequately the spatial and temporal changes in the lagoon (Figure 2). However, the fluorometry method (CARM) tends to produce lower concentrations in comparison with the spectrophotometric method (IEO-CSIC). For this reason, authors performed comparisons between satellite and *in situ* data independently for each database.

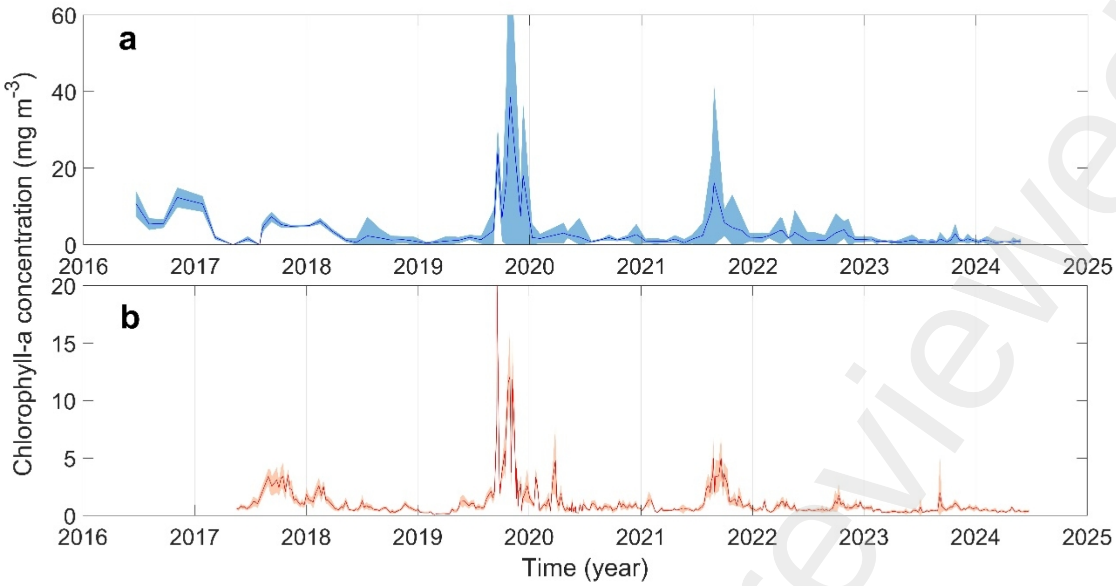

Figure 2. Time series of in situ *chlorophyll data based on spectrophotometer method (a; data from IEO-CSIC) and fluorometry (b; data from CARM cruises). The solid line corresponds to the average value of each sampling day for all the stations in the lagoon, and the shaded surface shows the 99% of the confidence interval of the values averaged for each sampling day.*

Every Level 2 satellite image covering all or part of the Mar Menor and nearby Mediterranean waters, captured by three hyperspectral sensors currently operational on six satellites from NASA's Ocean Color data service (https://oceancolor.gsfc.nasa.gov/) between 2000 and 2023 was downloaded (MODIS-Aqua, MODIS-Terra, VIIRS-SuomiNPP, VIIRS-NOAA20, OLCI-Sentinel-3A, and OLCI-Sentinel-3B). The main features of these sensors are shown in Table 1. Spectral reflectance data for specific locations within the lagoon were gathered from these images. The locations included the 6 sampling stations from the IEO-CSIC sampling program (labeled "A" in Figure 1), the 12 stations of the CARM program (labeled "B"), and 96 additional locations distributed over the whole lagoon at a spatial resolution of about 1 km (blue dots in Figure 1). Twenty-one additional locations in the Mediterranean Sea, outside of the lagoon, spaced at intervals of 1 km (labeled "E" in Figure 1) over or beyond the isobath of 30 meters were also selected (optically deep waters: McKinna & Werdell, 2018; IOCCG, 2000) in order to evaluate the influence of the lagoon bottom on the reflectance signal captured by the satellite. Altogether, time series of satellite reflectance data for 135 locations were gathered. Reflectance values for each location were obtained by linear interpolation from the nearest pixels of each scene that partially or fully covered the lagoon. Limited quality data due to presence of clouds and over land observations were discarded.

Table 1. Main features of the evaluated Ocean Color sensors.

Satellite/Sensor	Visible bands	Band Width	Revisit time	Equatorial crossing	Resolution at nadir	Start operation
TERRA/MODIS	10	10 nm	1-2d	10:30	1000 m	02/2000
AQUA/MODIS	10	10 nm	1-2d	13:30	1000 m	07/2002
SNPP/VIIRS	6	20 nm	1d	13:30	750 m	01/2012
NOAA-20/VIIRS	6	20 nm	1d	13:30	750 m	12/2017

S3-A/OLCI	10	10 nm	1d	10:00	300 m	05/2016
S3-B/OLCI	10	10 nm	1d	10:00	300 m	05/2018

A preliminary quality control was performed to clean the satellite database consisting of the analysis of reflectance values of each visible spectral band and the aerosol optical thickness (AOT) for each single sampling node. Observations with negative reflectance values in some spectral band or with AOT higher than 0.3 were discarded (Gómez-Jakobsen et al., 2018).

The distance to the coastline and the depth of each observation node were also used as a quality criterion. In particular, satellite information of nodes located at less than 2 km from the coastline or at depths shallower than 2 m were also discarded. This information was obtained from the data of coastline contour provided by the Instituto Hidrográfico de la Marina (IHM; https://ideihm.covam.es/portal/) and the bathymetry information generated from the own IEO-CSIC cruises (https://www.ieo.es/documents/10640/7712860/Informe_cientifico+tecnico_BATIMETRIA+Y+GEOLOGIA.pdf/a8fbcb92-c6a1-4367-aee8-c36b0386f289).

The process to design the algorithm consisted of three sequential steps. First, the reflectance spectra obtained after quality control for locations inside and outside the lagoon were compared to determine the influence of bottom reflectance on satellite-derived reflectance in the Mar Menor, especially in the shallower locations. For this, the satellite reflectance obtained during time periods featured by low and high Chla, according to the previously available information, were compared. These comparisons permitted the identification of reflectance bands affected by bottom reflection that should consequently be discarded for building the satellite empirical algorithm. This exercise also enabled the assessment of whether the bottom reflectivity signature changed following the first massive algal bloom in 2015, as changes in the bottom coverage have been observed since then (Belando et al., 2019).

Second, a non-hierarchical cluster analysis (k-means) was applied to the reflectance spectra of locations within the lagoon from 2002 to 2023, to identify periods characterized by distinct spectral signatures. To improve the analysis, the reflectance spectra were standardized by normalizing the Euclidean distance of each spectrum to unity. Standardized spectra were preferred over raw spectra to emphasize differences in spectral shape while minimizing the influence of absolute reflectance magnitudes, which can be affected by biases introduced during the standard atmospheric correction of each satellite scene. The optimal number of clusters was determined using two criteria: The Silhouette parameter (Rousseeuw, 1987) to maximize intra-cluster cohesion and inter-cluster separation, or an explained variance threshold of at least 80% as evaluated using the RS index (Halkidi et al., 2001). The normalized reflectance signatures and cluster centroids were subsequently analyzed to identify key spectral features associated with periods of varying Chla. For these analyses, MODIS and OLCI satellite imagery were utilized due to their well-balanced spectral resolution across the visible spectrum (400–

700 nm). VIIRS imagery was excluded from the analysis as its spectral coverage includes only a single band in the red region of the visible spectrum, limiting its suitability for detailed spectral assessments.

As a result of the initial steps, reflectance bands sensitive to variations in Chla concentrations within the lagoon were identified and utilized for the algorithm development. Satellite-derived reflectance values corresponding to spatial and temporal matches with the two *in situ* monitoring programs were extracted. Reflectance data were limited to observations acquired on the same day as *in situ* sampling, with aerosol optical thickness (AOT) values below 0.04, ensuring data quality. The *in situ* measurements were further constrained to conditions characterized by low suspended solid concentrations in the water column, typically occurring during episodic runoff events or sediment resuspension in shallow areas caused by strong winds. Polynomial functions were then fitted to establish the relationship between *in situ* Chla and satellite-derived reflectance expression. Two empirical algorithms were developed, leveraging fluorometric and spectrophotometric *in situ* Chla data. The satellite-derived chlorophyll concentrations were subsequently validated by comparison with the corresponding *in situ* measurements to assess the performance of the algorithms.

3. Results and Discussion

3.1. Reflectance characteristics of the Mar Menor and analysis of the available satellite data

Figure 3 shows the annual number of Level-2 satellite daily observations retained after quality control across various platforms over the entire study period in the Mar Menor. As anticipated, the availability of valid daily images decreases near the coastline. However, notable differences are observed among platforms. For instance, OLCI maintains a relatively high number of valid images at distances up to 600 m from the shoreline, whereas the performance of VIIRS and MODIS diminishes at distances below approximately 1.5 km and 2 km, respectively. These thresholds align closely with twice the spatial resolution of each sensor, a critical factor that must be considered when analyzing the temporal and spatial distribution patterns of satellite-derived chlorophyll data from specific platforms. Beyond the coastal influence, the maximum number of valid daily observations per year ranges from approximately 180 records for VIIRS to 100 records for OLCI. Based on these analyses, Sentinel-3 (OLCI) emerges as the most suitable sensor for mapping the spatial distribution of chlorophyll in the Mar Menor, although VIIRS exhibits the fewest temporal gaps, which makes it advantageous for time-series analyses.

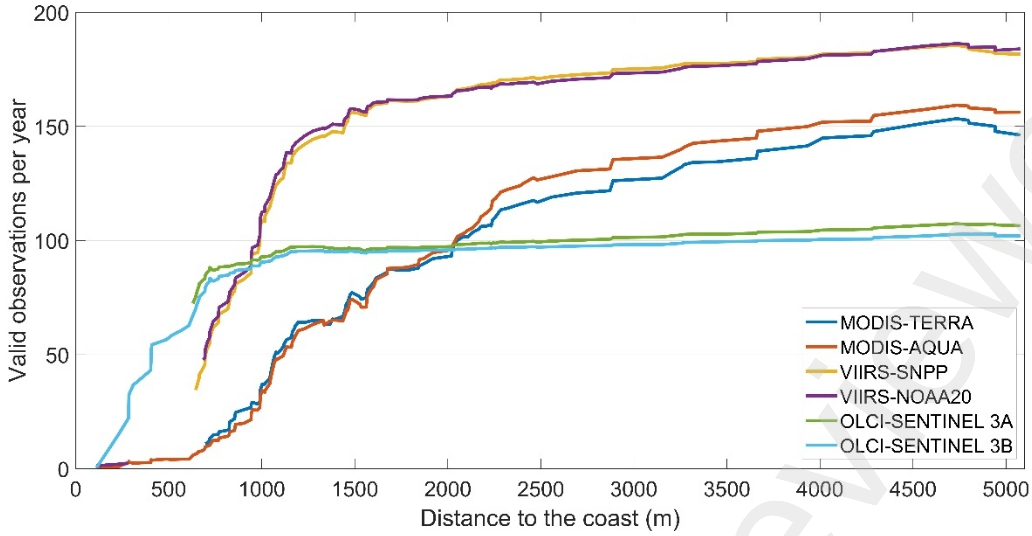

Figure 3. Average number of valid observations by year as a function of the distance to the coastline in the Mar Menor lagoon for the sensors evaluated.

Figure 4 shows the averaged reflectance spectra captured by the MODIS-AQUA sensor from the Mar Menor during two distinct time periods characterized by different trophic statuses (2000–2007 and 2014–2022). The spectra reveal that, within the lagoon, reflectance consistently peaked in the green band, centered around 550 nm, during both periods. This spectral signature contrasts with that observed outside the lagoon, in the Mediterranean Sea (stations E in Fig. 1), where maximum reflectance was recorded in the blue band. Despite these distinct spectral patterns between the lagoon's interior and exterior, temporal changes in the reflectance spectra were evident in both areas. Within the Mar Menor, a significant increase in overall reflectance was observed during the second period, with the most pronounced changes occurring in the green bands. Conversely, reflectance outside the lagoon showed a slight decline, particularly in the blue bands between 460 and 480 nm. This reduction likely reflects enhanced blue light absorption, indicative of increased chlorophyll concentrations, as previously reported by Gómez-Jakobsen et al. (2022) for this region.

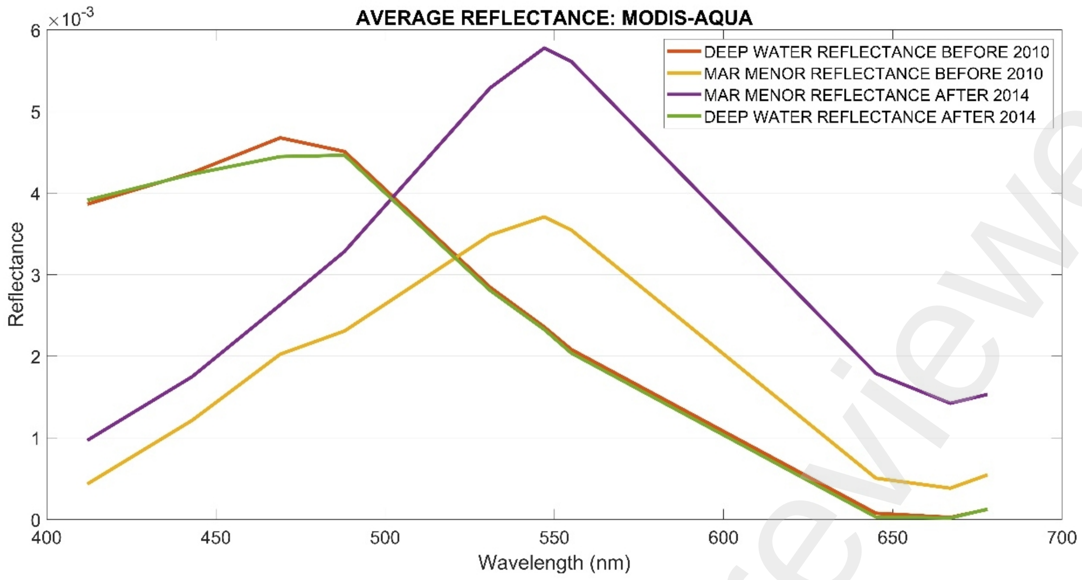

Figure 4. Average reflectance spectrum of the Mar Menor (shallow waters) and the neighboring Mediterranean waters (optically deep waters), for the periods before 2010, and after 2014.

3.2. Selection of spectral bands

Two independent non-hierarchical cluster analyses of normalized reflectance spectra were performed with all the available daily spectral reflectance data obtained from the locations within the lagoon by the MODIS-Aqua and OLCI-Sentinel3A sensors. For both sensors, the observations were grouped into three clusters according to the two statistical criteria shown in Table 2.

Table 2. Evolution of the Silhouette and RS indexes to determine the optimal number of clusters computed from the normalized spectra.

Sensor/Satellite	Index	2 clusters	**3 clusters**	4 clusters
MODIS-Aqua	Silhouette	0.59	0.48	0.47
	RS (%)	63	***80***	87
OLCI-Sentinel 3A	Silhouette	0.67	***0.49***	0.50
	RS (%)	69	***83***	90

The distribution of the available *in situ* Chla concentration values corresponding to the locations and dates of the reflectance spectra grouped into each cluster are shown in Figures 5b and 6b. For both platforms, the reflectance spectra grouped into cluster 1 come from locations and sampling dates featured by higher values of Chla concentration compared to those in clusters 2 and 3 (Figures 5a and 6a). Consequently, spectra or centroids assigned to cluster 1 show relatively higher values of reflectance in the red compared to the green band. The lower Chla concentrations associated with clusters 2 and 3 are reflected in the increased relative importance of reflectance in the deep blue bands (400–410 nm). The temporal distribution of the spectrum percentages assigned to each cluster, based on data from the selected locations, reveals that cluster 1 spectra dominate across the entire lagoon during periods characterized by elevated chlorophyll concentrations (Figures 5c and 6c). Cluster 2 exhibited lower relative reflectance in the deep blue bands compared to Cluster 3, which could be attributed to the absence of vegetation on the lagoon's bottom. If this hypothesis is valid, these periods of low Chla could be further analyzed to infer the extend of bottom vegetation coverage. Notably, the lagoon's bottom underwent re-colonization during the interval between the first phytoplankton bloom in 2016 and the subsequent bloom in 2019 (Belando et al., 2019, 2024; Jiménez-Casero, 2024). Our findings show that the temporal patterns in the reflectance spectra were consistent across MODIS and OLCI data, both of which share a similar spectral resolution (Figures 5c and 6c). In contrast, the cluster analysis performed using VIIRS normalized spectra did not yield clear results, likely due to the sensor's lower spectral resolution (data not shown). However, the algorithms developed should remain applicable to VIIRS, provided that this sensor offers data from the specific bands utilized in constructing the new algorithm.

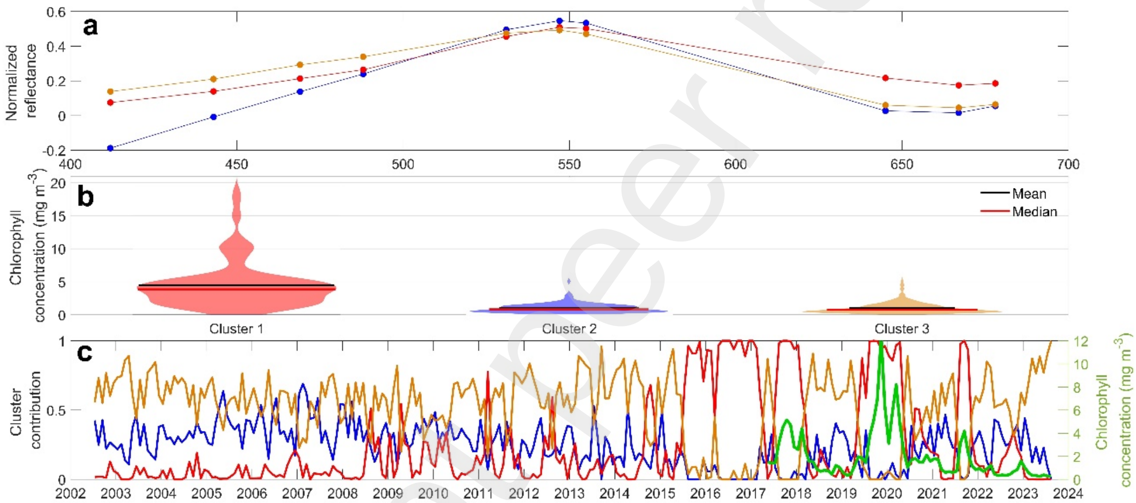

Figure 5. Resulting cluster analysis of the MODIS-Aqua reflectance normalized spectra of the nodes inside the Mar Menor lagoon, only using nodes with a minimum distance to the coastline of 2 km and located at depths of more than 2 meters, and satellite observations with AOT below 0.3. a) Centroids from normalized spectra, b) violin plot of in situ *$Chla_f$ matchups of each cluster, and c) temporal evolution of the proportion of the clusters.*

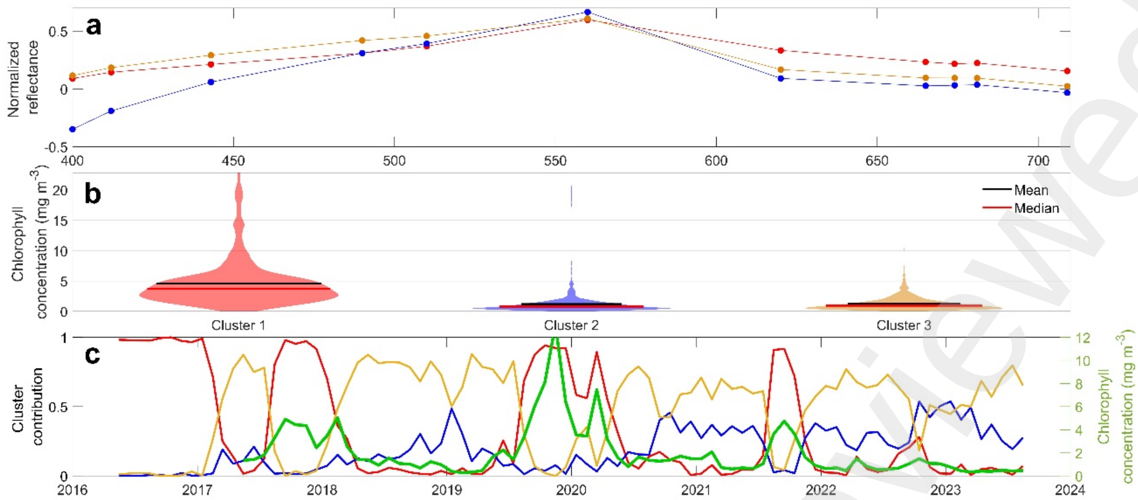

Figure 6. Resulting cluster analysis of the OLCI-Sentinel3A reflectance normalized spectra in the nodes inside the Mar Menor lagoon, only using nodes with a minimum distance to the coastline of 2 km and located at depths of more than 2 meters, and satellite observations with AOT below 0.3. a) Centroids from normalized spectra, b) violin plot of in situ *$Chla_f$ matchups of each cluster, and c) temporal evolution of the proportion of the clusters.*

According to the observed spectral signatures for the two periods featured by different chlorophyll levels, the red to green band ratio (RG) appears to be the most suitable to describe variations in chlorophyll (see Eq.1). Note that both spectral bands present a quite similar width in the 6 sensors analyzed in this work, with peaks centered around 550 nm and 670 nm (see Eq.1).

$$RG = Log_{10}(Rrs\_670/Rrs\_550) \quad Eq.1$$

3.3. BELA algorithms

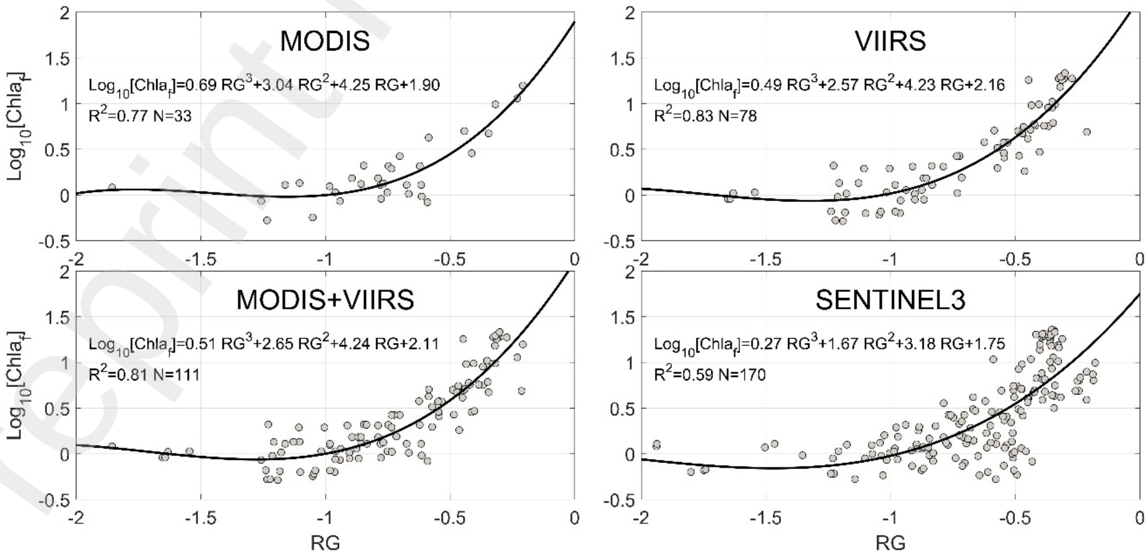

Figure 7. Fitting functions of the relationships between the RG index and in situ *chlorophyll* a *concentration (Chla) for the matchup dataset obtained with the fluorimetry concentration and the satellite data from the different analyzed sensors.*

All the matchups found inside the lagoon between *in situ* chlorophyll *a* based on fluorimetry and the RG band ratio were used to calculate a suitable expression for the satellite chlorophyll algorithm. As a first step, each sensor was analyzed separately (Figure 7). The best fit was obtained with a third order polynomial function (determination coefficients above 0.77 for most sensors). Note that at $\log_{10}$[Chla] values higher than -1, RG correlated with the *in situ* chlorophyll *a* concentration following a polynomial-like function. The intersect value of this function was about 2, and an RG value of -1 corresponds roughly to a value of $\log_{10}$[Chla] of 0 (i.e., 1 mg m^{-3} in linear scale). For RG band ratio values below -1, the correlation between $\log_{10}$[Chla] and RG was non-significant and, consequently, the algorithm does not perform effectively in this range of concentrations (below 2 mg m^{-3}). Coincidently, these conditions of low *in situ* chlorophyll concentrations likely correspond to time periods of high transparency, during which bottom reflectance can influence the reflectance captured by the sensor. Figure 7 also shows that the fit of the function is worse for Sentinel-3 compared with the +NASA sensors (MODIS+VIIRS), mainly due to a data group in the range of RG over -0.75 with $\log_{10}$[Chla] values lower than expected. This low performance of the RG index could be due to limitations of the OLCI sensors, as Gómez-Jakobsen et al. (2018) showed that the MERIS sensor (the predecessor of OLCI) presented a very strong dependence on aerosols in the blue-to-green band ratio. However, in our analysis, the aerosol importance must be quite limited as AOT values below 0.04 were selected for this calibration. Therefore, we suggest that these anomalous values could be related to other optical or observational factors of this sensor, which should be further investigated. Note that the cluster analysis for MODIS and OLCI revealed that high chlorophyll concentrations in the lagoon are directly associated with the red bands (RG index). This anomaly may be linked to the OLCI sensor or the Sentinel-3 orbits; Sentinel-3 captures images of the lagoon earlier in the morning (10 AM, as shown in Table 1) compared to other Ocean Color sensors, increasing the likelihood of glint in the images (unable to be detected by the Sentinel-3 Level-2 standard processing pipeline). The effect of the partial sun glint may be an increase in the red relative to green band reflectance, especially when atmospheric transparency is high. This fact, along with the finer spatial resolution of OLCI, suggests that the partial glint may explain why some RG values from OLCI fall outside the curve observed by other sensors. The anomalous data in the Sentinel-3 database are not equally distributed along the evaluated time period, increasing from 2019 to 2022. These anomalies are also not spatially homogeneous, being more frequent at certain latitudes or stations within the lagoon. In any case, these higher- than-expected RG values should be detected and removed to refine the algorithm for OLCI data.

Since the fitting functions between *in situ* chlorophyll estimated with fluorometry probe and RG are quite similar for the different platforms analyzed, we propose using the expression shown in Figure 8 and Eq. 2 (BELA algorithm) to produce values of satellite chlorophyll concentration from the RG index, irrespectively of the satellite platform from which the data are retrieved. Note that the Eq. 2 was derived after removing the anomalous data commented above from the Sentinel-3 observations. For this, the NASA matchup dataset was analyzed and RG data out of the 95% confidence interval were discarded (approx. 16% of the registers for AOT<0.04; Fig. 8). Interestingly, when the quality requirements for AOT are relaxed by including matchups with AOT values below 0.1, the number of matchups increases, and the percentage of anomalous data decreases to 7%. This suggests that the amount of anomalous RG values is proportional to atmospheric transparency, which supports the partial glint hypothesis. Furthermore, the shape of the fitting function does not change substantially when the RG index values coming from images with AOT lower than 0.1 are included (although goodness of the fit is worse, R^2= 0.67). However, the intersection point with the y-axis decreases to 1.7 because the number of matchups with high *in situ* chlorophyll concentration also decreases.

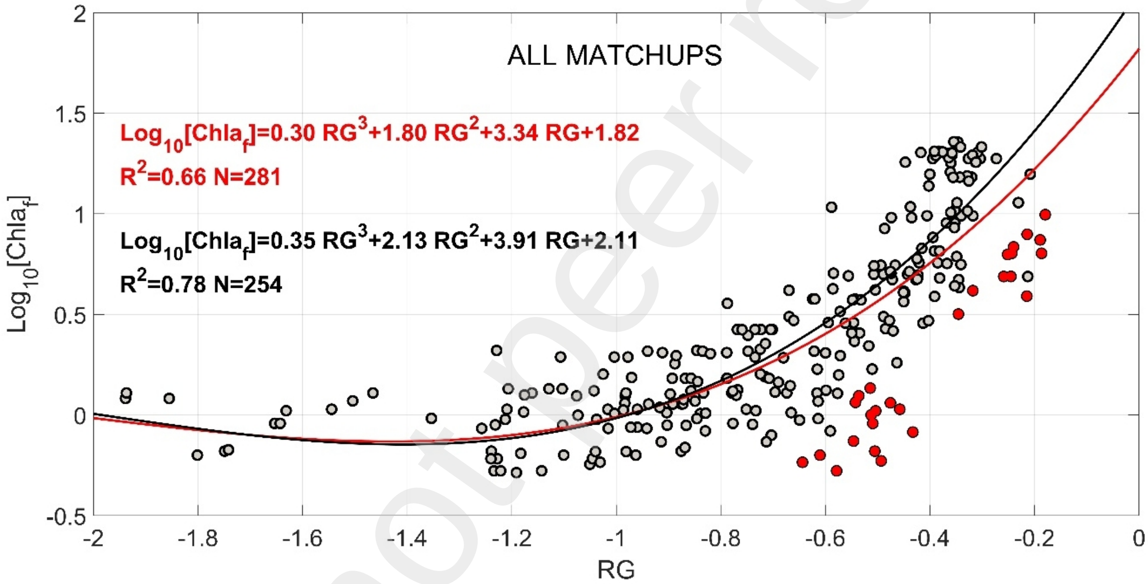

Figure 8. Fitting function of RG to the in situ *chlorophyll concentrations based on* in situ *fluorimetry probe ($Chla_f$), discarding only those Sentinel-3 matchups falling down of the 95 confidence interval of the NASA matchups (red circles).*

$$[Chla]fluorometry = 0.353RG^3 + 2.132RG^2 + 3.905RG + 2.110 \quad Eq.2$$

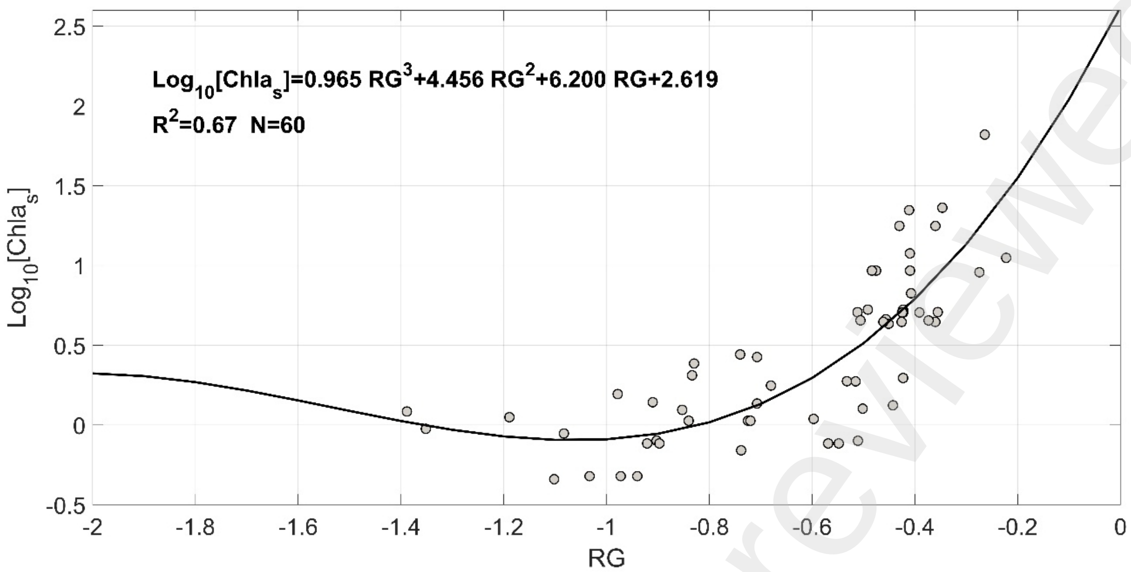

Figure 9. Fit function of RG to the in situ *chlorophyll concentrations based on spectrophotometric method ($Chla_s$).*

$$[Chla]spectrophotometry = 0.965RG^3 + 4.456RG^2 + 6.200RG + 2.619 \quad Eq.3$$

The effect of using chlorophyll data obtained with the spectrophotometric method on the empirical algorithm is illustrated in Fig. 9. The intercept of this function (2.62) is somewhat higher than that obtained for the function based on fluorometry data, and the estimated value of *log10[Chla]* at *RG=-1* is slightly lower than 0. Nevertheless, the spectrophotometric data were obtained at 4 m depth while the fluorometry data were computed from the vertical profiles. In order to determine if this different vertical resolution accounts for the slight differences in the shape of both fitting functions, the relationship between the fluorometry chlorophyll concentration at 4 m depth and the vertical average of these profiles was analyzed (Fig.10a). The determination coefficient of this correlation and the slope of the linear regression were 0.996 and 1.27, respectively (R^2 was calculated by applying the bi-squared robust scheme). The correlations of the vertical average values with values obtained at other specific depths are presented in Table 3.

Table 3. Parameters of the linear regressions between the chlorophyll values at different depths and the corresponding vertical average in the Mar Menor lagoon, calculated from the fluorescence profiles ($Chla_f$).

		BISQUARED			STANDARD		
Depth (m)	N	Slope	R^2	Intercept	Slope	R^2	Intercept
1	4231	0.723	0.995	0.059	0.804	0.862	-0.087
2	4029	1.043	0.998	-0.027	1.131	0.946	-0.165
3	3698	1.178	0.998	-0.034	1.145	0.946	0.038
4	***3617***	***1.274***	***0.996***	***0.021***	***1.104***	***0.875***	***0.232***

5	3273	1.195	0.991	0.076	1.093	0.829	0.341

These significant relationships suggest that the sampling depth is not a major factor in affecting the performance of the satellite algorithm, provided that the water column in Mar Menor is typically well-mixed vertically. In fact, if the data of chlorophyll concentration obtained with the algorithm based on fluorometry data are used to calculate the satellite concentration at 4 m depth (Fig. 10a), the correlation between the data pools of satellite chlorophyll derived from the two algorithms is highly significant (Fig. 10b; R^2=0.90), and the slope of this linear regression is close to the unity (1.025). However, at high chlorophyll concentrations (up 30 mg m^{-3}), the algorithm based on spectrophotometric data produces higher values, showing an intercept coefficient of -1.495 mg m^{-3}. These comparisons indicate that the effect of the method used to obtain *in situ* chlorophyll values is more pronounced at higher concentrations. Such differences may arise from uneven vertical distribution of phytoplankton, especially when increasing cell abundance tends to form aggregates. However, it has been demonstrated that the relationship between the concentration of Chla and the intensity of its fluorescence becomes non-linear depending on the spectral composition of the underwater irradiance and/or changes in the composition or physiological status of the phytoplankton (e.g., Ostrowska et al. 2015). Therefore, the algorithm based on fluorometry data would have a worse performance at high concentrations of chlorophyll in the lagoon in comparison to the algorithm based on spectrophotometric data.

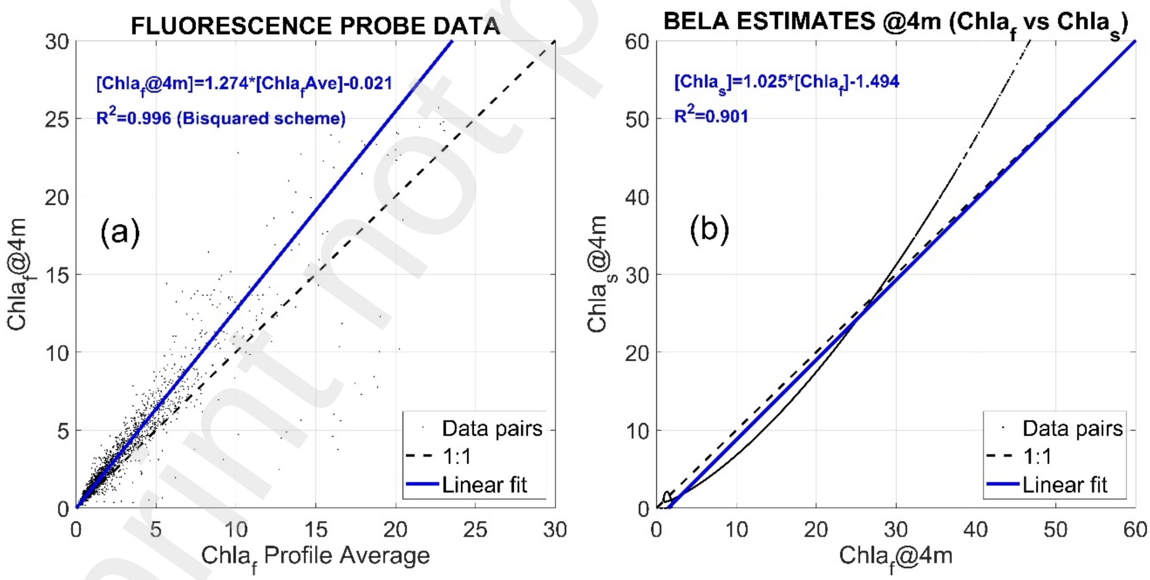

Figure 10. Comparisons of chlorophyll measurements: a) the vertical average versus the value registered at 4 meters from the fluorescence probe database; b) Estimated chlorophyll data with the BELA algorithms fitted for the fluorescence probe and spectrophotometric databases at 4 meters.

3.3. Performance of the BELA algorithms in producing time series of Chla for the lagoon

The performance of the two BELA algorithms in producing comparable values of Chla is illustrated in Fig. 11. The time series shown was computed as the daily averages of the *in situ* chlorophyll *a* concentrations and the average from the different satellite estimates available in all the nodes inside the lagoon for a given day. Only the days accounting for a minimum valid spatial data over the lagoon are represented to avoid biases

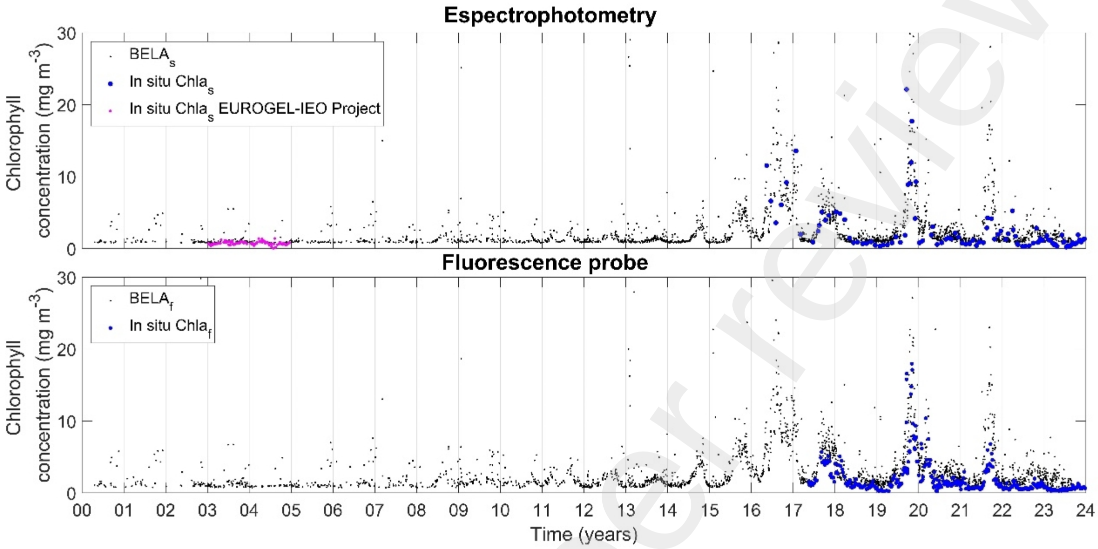

Figure 11. Representation of the daily averaged chlorophyll concentration in the Mar Menor waters estimated by means of BELA algorithms versus the in situ daily averaged chlorophyll concentration. Versions for spectrophotometry data (upper panel) and in situ *fluorometry probe data (lower panel) databases. We also presented daily averaged chlorophyll data from the project EUROGEL-IEO between 2003 and 2004, from weekly samplings in 24 sampling stations distributed in the Mar Menor lagoon (also measured by the spectrophotometric method).*

The algorithms BELA is not able to detect changes in chlorophyll in the lower range of concentrations in the Mar Menor lagoon (below 2 mg m^{-3}), due to the important reflectance of the shallow bottom. However, it is less dependent on the bottom reflectance than algorithms based on the blue or green wavelengths, as the red light penetrate less in the water column. Furthermore, the BELA algorithms are able to detect the values of chlorophyll related to abnormal productivity phenomena in the lagoon in a wide range of concentrations, mainly above 2 mg/m^3. The estimations based on the BELA algorithms showed that the variability of the chlorophyll observed during the first years of the 2000's was low, as historically observed in the lagoon (Figure 11). Between 2008 and 2014, the intra-annual variability of averaged chlorophyll concentrations increased compared to earlier years, indicating more frequent periods of higher productivity. Since 2015, periods of very high chlorophyll concentrations have been clearly observable in the estimated time series. Based on these patterns, three main periods can be identified since 2000 from the chlorophyll time series. Additionally, it was noted that, in general, the annual primary chlorophyll peaks occurred during the second half of the year, typically beginning in early autumn and persisting until late winter or early spring, with varying durations depending on the year. This marked seasonality is evident in the data and reflects recurring patterns of chlorophyll dynamics in the lagoon. Spatial analysis of the observed blooms further highlights the distribution and intensity of nutrient-driven productivity across the lagoon.

The algorithms proposed in this study are applicable to all the sensors tested, as well as to other similar or upcoming sensors that share the same spectral features, for example VIIRS on board NOAA-21 or OCI on board PACE. This versatility of the designed algorithms allows to increase the days with valid observations, as the different ocean color platforms scan each location on the lagoon's surface at different times of the day and under varying observation geometries, thereby increasing the useful available information over time (Giménez et al., 2024). Additionally, the potential use of combined data from several ocean color sensors using the BELA algorithms can improve the estimated values for each single day and location, providing reliable data for the monitoring and assessment of the state of the Mar Menor lagoon in the future. For example, we showed that OLCI-Sentinel-3 Level 2 observations hosted in the NASA Ocean Color web page could be biased by the partial effect of the sun glint. In the operational use of these algorithms, we propose to detect this effect from the observations of the other ocean color sensors on the same days and nearby locations. Note that since June of 2018, when the 6 evaluated sensors were operative, approximately 50% of the daily chlorophyll values came from at least 2 satellite observations for each node. For the same time period, we produced valid data in each single node for about 34% of all the days.

4. Conclusions

Two algorithms ($BELA_f$ and $BELA_s$) are proposed for estimating Chla concentrations in the Mar Menor coastal lagoon, based on the comparison between satellite-derived red and green band reflectance data and *in situ* chlorophyll concentrations obtained through fluorometry ($BELA_f$) or spectrophotometry ($BELA_s$). Both algorithms demonstrate robust performance for *in situ* chlorophyll concentrations exceeding 2 mg m^{-3}. However, at concentrations below this threshold, reflectance signals captured by satellites are influenced by bottom reflectivity, reducing accuracy. The BELA algorithms are suitable for concentrations up to 30 mg m^{-3}, but at higher levels, the $BELA_f$ algorithm, based on fluorometry, underestimates *in situ* concentrations, likely due to the limitations of fluorescence as a proxy for chlorophyll estimation.

Cluster analysis further suggest that ocean color sensors have the potential to provide insights into the vegetation coverage of the lagoon bottom. We propose exploring the ratio of deep blue (approximately 400–410 nm) to green bands for this purpose, particularly in cases where the BELA algorithms identify clear waters, to assess the presence or absence of bottom vegetation. Among the current ocean color sensors, OLCI is particularly recommended due to its higher spatial resolution, enabling the generation of detailed maps to monitor the evolution of vegetation coverage over time.

BELA algorithms will permit detecting short-term changes in the productivity of the lagoon and will be a useful tool to early warning of phytoplankton bloom events. Additionally, BELA will be usable to build a long time series of chlorophyll which will provide information about the past ecological status of the lagoon and the success of the actions introduced aiming to achieve its re-oligotrophication.

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